DigiLocker Issuer Application

Production Technical Design Document (Final v1.1)

1. Objective

Build a **DLTS-compliant Issuer service** that:

- Implements **Pull URI API (only API)**
- Returns:
  - URI
  - Base64 PDF
  - Base64 XML metadata
- Enforces:
  - HMAC authentication
  - Identity validation (post-lookup)
  - Document integrity

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2. Spec Compliance (Verified)

From v1.13:

- ✓ Only **Pull URI API** exists
- ✓ PullDoc removed
- ✓ Response includes document content
- ✓ URI format mandatory
- ✓ Identity validation recommended

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3. System Architecture

```
[DigiLocker]
    |
```

```
    v
[API Layer]
   ├── HMAC Validator
   ├── XML Parser
   ├── Request Validator
   ├── Document Service
   ├── URI Service
   ├── File Service
   ├── Response Builder
   ├── Audit Logger
    |
    v
[PostgreSQL] + [File Storage]
```

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4. Configuration

```
ISSUER_ID=in.gov.state.department
API_KEY=<shared-secret>

BASE_STORAGE_PATH=/data/docs

INTEGRITY_MODE=STRICT | LOG_AND_SERVE
IDENTITY_VALIDATION_MODE=STRICT | LENIENT

MAX_FILE_SIZE_MB=10
REQUEST_TIMEOUT_MS=5000
```

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5. Data Model

5.1 Table: documents

```
CREATE TABLE documents (
  id BIGSERIAL PRIMARY KEY,

  authorization_number VARCHAR(50) NOT NULL,
  document_type VARCHAR(5) NOT NULL,

  doc_id VARCHAR(10) UNIQUE,
  uri VARCHAR(255) UNIQUE,

  employee_name TEXT,
  employee_dob DATE,

  file_relative_path TEXT NOT NULL,
  file_checksum TEXT,

  is_active BOOLEAN DEFAULT TRUE,

  created_at TIMESTAMP,
  updated_at TIMESTAMP
);
```

5.2 Constraints

UNIQUE (authorization_number, document_type)

```
CHECK (document_type ~ '^[A-Za-z]{1,5}$')
```

5.3 Indexes

```
CREATE INDEX idx_lookup
ON documents (authorization_number, document_type);

CREATE INDEX idx_uri ON documents (uri);
```

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6. URI Management

6.1 Format

URI = ISSUER_ID + "-" + document_type + "-" + doc_id

6.2 Strategy

- Lazy generation
- Atomic
- Persistent

6.3 Algorithm

```
def ensure_uri(row_id):
    with transaction():
        row = SELECT ... FOR UPDATE

        if row.uri:
            return row.uri
```

```
doc_id = random_string(10)
uri = f"{ISSUER_ID}-{row.document_type}-{doc_id}"

UPDATE documents
SET doc_id = doc_id, uri = uri
WHERE id = row_id

return uri
```

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7. Authentication

7.1 HMAC Validation

expected = Base64(HMAC_SHA256(request_body, API_KEY))

Compare with:

x-digilocker-hmac

7.2 KeyHash Validation

keyhash = SHA256(API_KEY + timestamp)

7.3 Rejection Rules

Condition	Action
HMAC mismatch	401
keyhash mismatch	401
timestamp invalid	401

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8. Request Handling Pipeline

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1. Parse XML
2. Validate HMAC
3. Validate keyhash + timestamp
4. Extract:
   - DocType
   - authorization_number (UDF)
   - DOB (optional)
   - Name (optional)
5. Lookup document
6. Identity validation
7. Ensure URI
8. File read + integrity check
9. Base64 encode
10. Build response XML
11. Log
12. Return

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9. Lookup vs Identity Validation (Final)

9.1 Lookup (Primary Key Resolution)

```
SELECT *
FROM documents
WHERE authorization_number = ?
AND document_type = ?
```

AND is_active = true;

9.2 Identity Validation (Access Control)

Rules

Field	Rule
DOB	exact match (if provided)
Name	normalized match (if provided)

Normalization

```
def normalize(name):
    return lower(remove_spaces(remove_punctuation(name)))
```

9.3 Mode

IDENTITY_VALIDATION_MODE=STRICT | LENIENT

STRICT

- Require at least one identity field
- Reject if mismatch

LENIENT

- Allow access without identity fields
- Still validate if provided

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10. File Handling

10.1 Path Resolution

path = BASE_STORAGE_PATH + file_relative_path

10.2 Integrity Check

Check	Action
file missing	fail
checksum mismatch	config-based

10.3 Modes

INTEGRITY_MODE=STRICT | LOG_AND_SERVE

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11. Response Construction

11.1 XML Structure

```
<PullURIResponse>
  <ResponseStatus Status="1" ts="..." txn="..." />
  <DocDetails>
    <IssuedTo>
      <Persons>
        <Person name="" dob="" />
      </Persons>
    </IssuedTo>
    <URI>...</URI>
    <DocContent>BASE64 PDF</DocContent>
```

```
  <DataContent>BASE64 XML</DataContent>
 </DocDetails>
</PullURIResponse>
```

11.2 Rules

- DocContent mandatory (pdf/both)
- DataContent mandatory
- synchronous response

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12. Error Handling

12.1 Response Status

Status	Meaning
1	success
0	failure
9	pending

12.2 Mapping

Scenario	Output
no record	Status=0
identity mismatch	Status=0
auth failure	HTTP 401
integrity fail	depends

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13. Security Model

13.1 Layers

Layer	Mechanism
Transport	HMAC
Request	keyhash
Access	identity validation

13.2 Prohibitions

- ❌ No public doc endpoint
- ❌ No URI-only access

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14. Logging & Audit

14.1 Access Log

- txn
- timestamp
- docType
- status
- latency

14.2 Integrity Log

- checksum mismatch
- file missing

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15. Performance

15.1 Constraints

- Base64 → +33% size
- Full file in memory

15.2 Limits

MAX_FILE_SIZE_MB=10

15.3 Concurrency

- bounded worker pool
- DB pool capped

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16. Background Jobs

16.1 URI Pre-generation

assign URI for documents where uri IS NULL

16.2 Integrity Scan

periodic checksum verification

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17. Deployment

Nginx

↓

App (Uvicorn/Gunicorn)

↓

PostgreSQL

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File Storage

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18. Observability

Metrics

- request rate
- error rate
- latency
- integrity failures

Alerts

- HMAC failures spike
- integrity failures
- DB latency

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19. Acceptance Criteria

✓ HMAC validation correct
✓ URI stable and unique
✓ No duplicate URI under concurrency
✓ Identity mismatch blocks access (STRICT mode)
✓ Response matches XML schema
✓ Integrity checks enforced